Communication Systems Lecture-4: Sampling Theorem

Chadi Abou-Rjeily

Department of Electrical and Computer Engineering
Lebanese American University
chadi.abourjeily@lau.edu.lb

September 5, 2024

Bandlimited Waveforms

- A waveform $w(t)$ is said to be (absolutely) bandlimited to B Hz if:

$$W(f) = \mathcal{F}[w(t)] = 0 \quad \text{for} \quad |f| \geq B$$

- A waveform $w(t)$ is said to be (absolutely) time limited if:

$$w(t) = 0 \quad \text{for} \quad |t| > T$$

- **Theorem**: An absolutely bandlimited waveform can not be absolutely time limited, and vice versa.
  - Example: $w(t) = \mathrm{rect}(t)$ with $W(f) = \mathrm{sinc}(f)$ is time limited and not bandlimited.

Parseval's theorem

Parseval's theorem [general form]:

$$\int_{-\infty}^{+\infty} w_1(t) w_2^*(t) \mathrm{d}t = \int_{-\infty}^{+\infty} W_1(f) W_2^*(f) \mathrm{d}f$$

where $W_1(f) = \mathcal{F}[w_1(t)]$ and $W_2(f) = \mathcal{F}[w_2(t)]$.

- If $w_1(t) = w_2(t) = w(t)$, then the theorem reduces to:

$$\int_{-\infty}^{+\infty} |w(t)|^2 \mathrm{d}t = \int_{-\infty}^{+\infty} |W(f)|^2 \mathrm{d}f$$

Sampling Theorem (1)

Any physical waveform may be represented over the interval $-\infty < t < +\infty$ by:

$$w(t) = \sum_{n=-\infty}^{+\infty} a_n \mathrm{sinc}\left\{ f_s \left[t - \left(\frac{n}{f_s} \right) \right] \right\}$$

where:

$$a_n = f_s \int_{-\infty}^{+\infty} w(t) \mathrm{sinc}\left\{ f_s \left[t - \left(\frac{n}{f_s} \right) \right] \right\} \mathrm{d}t$$

where f_s is a parameter that is assigned some convenient value greater than zero.

Proof:

- The sampling theorem can be seen as an application of the orthogonal series representation of signals.
- We need to show that:

$$\phi_n(t) = \mathrm{sinc}\left\{ f_s \left[t - \left(\frac{n}{f_s} \right) \right] \right\} \quad ; \quad n \in \mathbb{Z}$$

form a set of orthogonal functions.

Sampling Theorem (2)

- In other words, we must show that:

$$\int_{-\infty}^{+\infty} \phi_n(t)\phi_m^*(t)\mathrm{d}t = K_n\delta_{nm}$$

- Applying Parseval's theorem:

$$\int_{-\infty}^{+\infty} \phi_n(t)\phi_m^*(t)\mathrm{d}t = \int_{-\infty}^{+\infty} \Phi_n(f)\Phi_m^*(f)\mathrm{d}f$$

where:

$$\Phi_n(f) = \mathcal{F}\left[\phi_n(t)\right] = \frac{1}{f_s}\mathrm{rect}\left(\frac{f}{f_s}\right)e^{-j2\pi(nf/f_s)}$$

- Consequently:

$$\int_{-\infty}^{+\infty} \phi_n(t)\phi_m^*(t)\mathrm{d}t = \frac{1}{f_s^2}\int_{-\infty}^{+\infty} \mathrm{rect}^2\left(\frac{f}{f_s}\right)e^{-j2\pi(nf/f_s)}e^{j2\pi(mf/f_s)}\mathrm{d}f$$

$$= \frac{1}{f_s^2}\int_{-f_s/2}^{+f_s/2} e^{-j2\pi((n-m)f/f_s)}\mathrm{d}f$$

Sampling Theorem (3)

- For $n = m$, $e^{-j2\pi((n-m)f/f_s)} = 1$ and:

$$\int_{-\infty}^{+\infty} \phi_n(t)\phi_m^*(t)\mathrm{d}t = \frac{1}{f_s^2}\int_{-f_s/2}^{+f_s/2} 1\mathrm{d}f = \frac{1}{f_s}$$

  implying that $K_n = \frac{1}{f_s}$.

- For $n \neq m$:

$$\begin{aligned}
\int_{-\infty}^{+\infty} \phi_n(t)\phi_m^*(t)\mathrm{d}t &= \frac{1}{f_s^2}\left[\frac{-e^{-j2\pi((n-m)f/f_s)}}{j2\pi(n-m)/f_s}\right]_{-f_s/2}^{f_s/2} \\
&= \frac{1}{f_s^2}\frac{e^{j\pi(n-m)} - e^{-j\pi(n-m)}}{j2\pi(n-m)/f_s} \\
&= \frac{1}{f_s}\frac{\sin(\pi(n-m))}{\pi(n-m)} \\
&= 0
\end{aligned}$$

Sampling Theorem (4)

- Consequently, the functions in the set

$$\left\{\phi_n = \mathrm{sinc}\left[f_s\left[t - \left(\frac{n}{f_s}\right)\right]\right]\right\}_{n=-\infty}^{+\infty}$$

  are orthogonal.
- Based on the orthogonal series representation of signals, the coefficients a_n can be calculated from:

$$\begin{aligned} a_n &= \frac{1}{K_n}\int_{-\infty}^{+\infty} w(t)\phi_n^*(t)\mathrm{d}t \\ &= f_s\int_{-\infty}^{+\infty} w(t)\mathrm{sinc}\left\{f_s\left[t - \left(\frac{n}{f_s}\right)\right]\right\}\mathrm{d}t \end{aligned}$$

Sampling Theorem (5)

Proposition: If $w(t)$ is absolutely bandlimited to B Hz with $f_s \geq 2B$, then the orthogonal series coefficients a_n are equal to the samples of $w(t)$ taken at the time instants n/f_s:

$$a_n = w\left(\frac{n}{f_s}\right)$$

Proof:

- The coefficients a_n can be calculated from:

$$a_n = \frac{1}{K_n}\int_{-\infty}^{+\infty} w(t)\phi_n^*(t)\mathrm{d}t = f_s\int_{-\infty}^{+\infty} w(t)\phi_n^*(t)\mathrm{d}t$$

- Applying Parseval's theorem:

$$a_n = f_s\int_{-\infty}^{+\infty} W(f)\Phi_n^*(f)\mathrm{d}f$$

where:

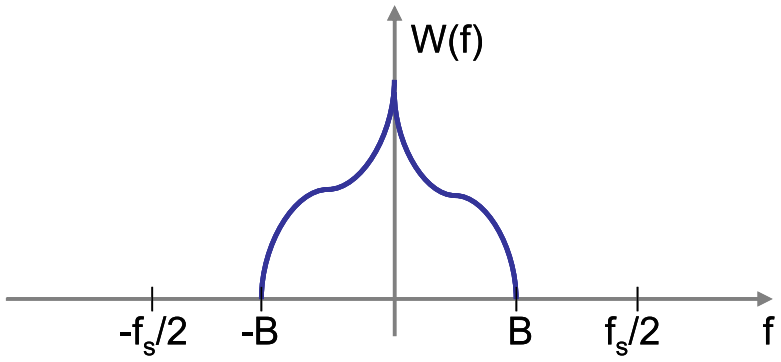
$$\Phi_n(f) = \frac{1}{f_s}\mathrm{rect}\left(\frac{f}{f_s}\right)e^{-j2\pi(nf/f_s)}$$

Sampling Theorem (6)

- Consequently:

$$a_n = \int_{-\infty}^{+\infty} W(f)\mathrm{rect}\left(\frac{f}{f_s}\right) e^{j2\pi(nf/f_s)}\mathrm{d}f$$
$$= \int_{-f_s/2}^{+f_s/2} W(f)e^{j2\pi(nf/f_s)}\mathrm{d}f$$

- Given that $W(f) = 0$ for $|f| > B$ and $f_s \geq 2B$ (implying that $B \leq f_s/2$), then the limits of the last integral may be extended to $(-\infty, +\infty)$.



Sampling Theorem (7)

- Finally:

$$a_n = \int_{-\infty}^{+\infty} W(f) e^{j2\pi(nf/f_s)} \mathrm{d}f$$

- On the other hand, $w(t)$ is the inverse Fourier transform of $W(f)$ and can be written as:

$$w(t) = \int_{-\infty}^{+\infty} W(f) e^{j2\pi ft} \mathrm{d}f$$

- Comparing the above equations shows that a_n is nothing but the inverse Fourier transform of $W(f)$ evaluated at the time instant $t = n/f_s$:

$$a_n = w\left(\frac{n}{f_s}\right)$$

Sampling Theorem (8)

- From what preceded, we conclude that the minimum sampling rate needed for the reconstruction of a bandlimited signal is:

$$\min[f_s] = 2B$$

  where B is the absolute bandwidth of the signal.
- The sampling frequency of $f_s = 2B$ is known as the Nyquist frequency.

Dimensionality Theorem (1)

In what follows, we are interested in reproducing a signal $w(t)$ over an interval of T_0 seconds.

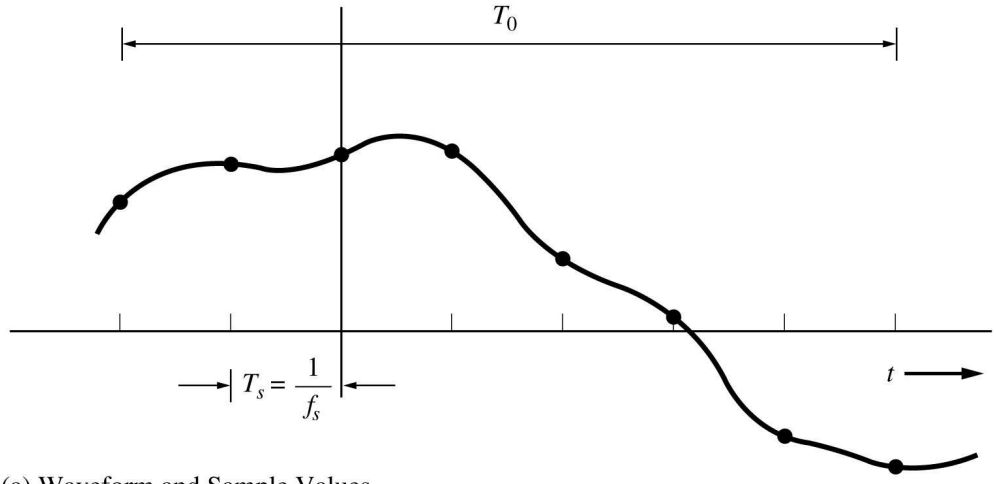
- Instead of having an infinite number of terms in the orthogonal series representation, $w(t)$ is approximated by:

$$w(t) \approx \sum_{n=n_1}^{n_1+N} a_n \phi_n(t)$$

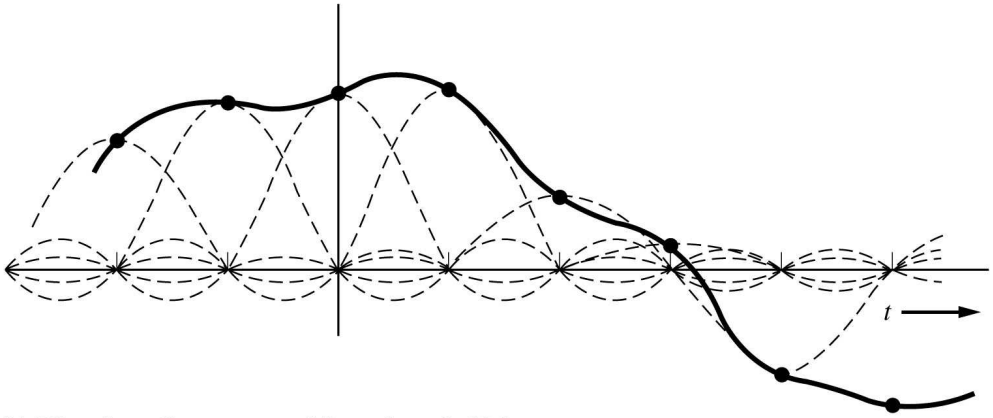
- In other words, over the interval of interest, $w(t)$ is <u>approximately</u> reconstituted by using N samples.

Dimensionality Theorem (2)

- The reconstructed waveform is obtained by the weighted sum of N time-delayed $\mathrm{sinc}(x)$ waveforms.
  - The weights are the sample values: $a_n = w(n/f_s)$.



(a) Waveform and Sample Values



(b) Waveform Reconstructed from Sample Values

Dimensionality Theorem (3)

- The minimum number of samples needed to reconstruct $w(t)$ over a duration of T_0 is:

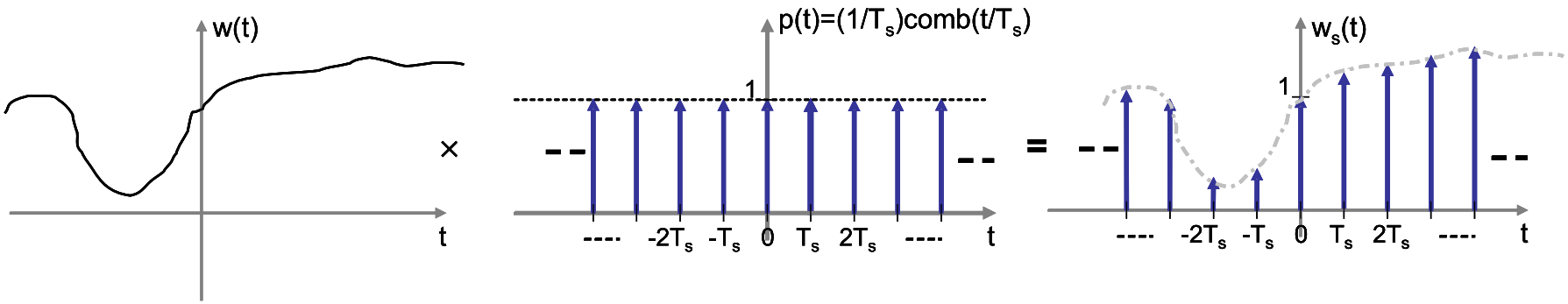
$$N = \frac{T_0}{1/f_s} = T_0 f_s \geq 2BT_0$$

  since $f_s \geq 2B$.

- **Dimensionality Theorem:** When BT_0 is large, a real waveform having an absolute bandwidth of B may be completely specified by $N = 2BT_0$ independent pieces of information that will describe the waveform over a T_0 interval.

  N is said to be the number of dimensions required to specify the waveform.

Impulse Sampling (1)



- Impulse sampling can be performed by multiplying the signal $w(t)$ by the sampling function $p(t)$:

$$
\begin{aligned}
w_s(t) &\triangleq w(t)p(t) \\
&= w(t)\sum_{n\in\mathbb{Z}} \delta(t - nT_s) \\
&= \sum_{n\in\mathbb{Z}} w(nT_s)\delta(t - nT_s)
\end{aligned}
$$

where $T_s = 1/f_s$.

Impulse Sampling (2)

- Given that $p(t)$ is periodic (with period T_s), it admits the following Fourier series representation:

$$p(t) = \sum_{n \in \mathbb{Z}} a_n e^{jn2\pi f_s t}$$

- The coefficients a_n can be calculated from:

$$a_n = \frac{1}{T_s} \int_{T_s} p(t) e^{-jn2\pi f_s t} \mathrm{d}t = \frac{1}{T_s} \int_{-T_s/2}^{T_s/2} p(t) e^{-jn2\pi f_s t} \mathrm{d}t$$
$$= \frac{1}{T_s} \int_{-T_s/2}^{T_s/2} \delta(t) e^{-jn2\pi f_s t} \mathrm{d}t = \frac{1}{T_s} \int_{-T_s/2}^{T_s/2} \delta(t) \mathrm{d}t = \frac{1}{T_s}$$

- Consequently:

$$p(t) = \frac{1}{T_s} \sum_{n \in \mathbb{Z}} e^{jn2\pi f_s t}$$
$$\Rightarrow \quad P(f) = \frac{1}{T_s} \sum_{n \in \mathbb{Z}} \delta(f - nf_s)$$

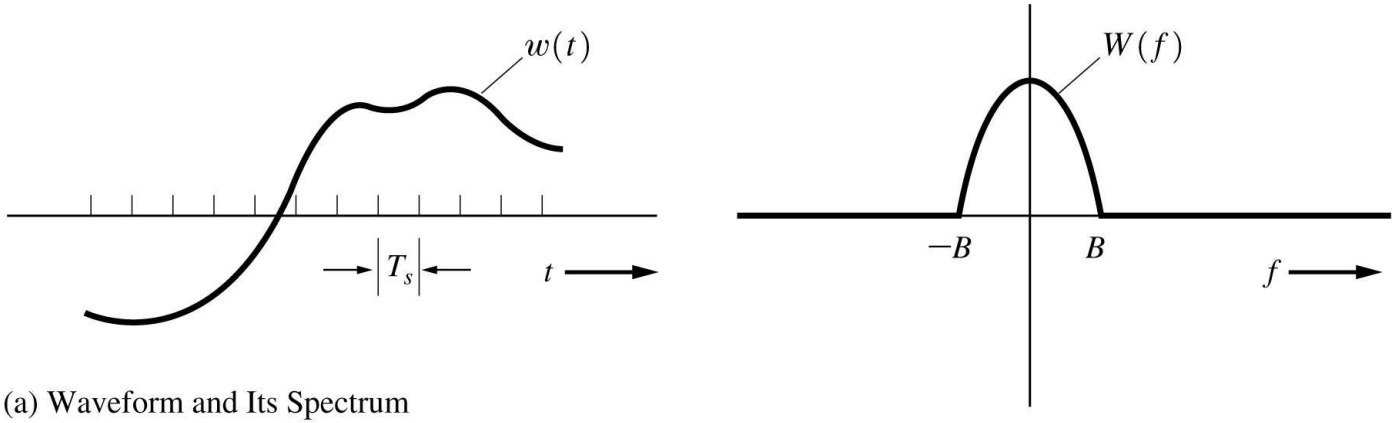
Impulse Sampling (3)

- Since $w_s(t) = w(t)p(t)$:

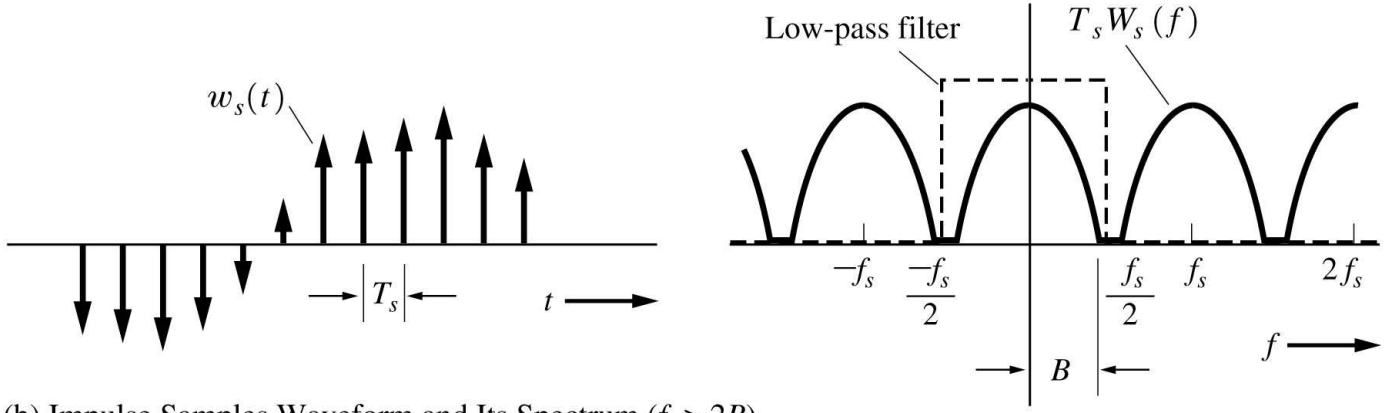
$$
\begin{aligned}
W_s(f) &= W(f) * P(f) \\
&= W(f) * \frac{1}{T_s} \sum_{n \in \mathbb{Z}} \delta(f - nf_s) \\
&= \frac{1}{T_s} \sum_{n \in \mathbb{Z}} W(f) * \delta(f - nf_s) \\
&= \frac{1}{T_s} \sum_{n \in \mathbb{Z}} W(f - nf_s)
\end{aligned}
$$

Impulse Sampling (4)

- The overlap between the shifted replicas of $W(f)$ can be avoided by choosing $f_s - B \geq B$ resulting in $f_s \geq 2B$.
- A low-pass filter whose cutoff frequency verifies $B \leq f_c \leq f_s - B$ can be used to recover $w(t)$.



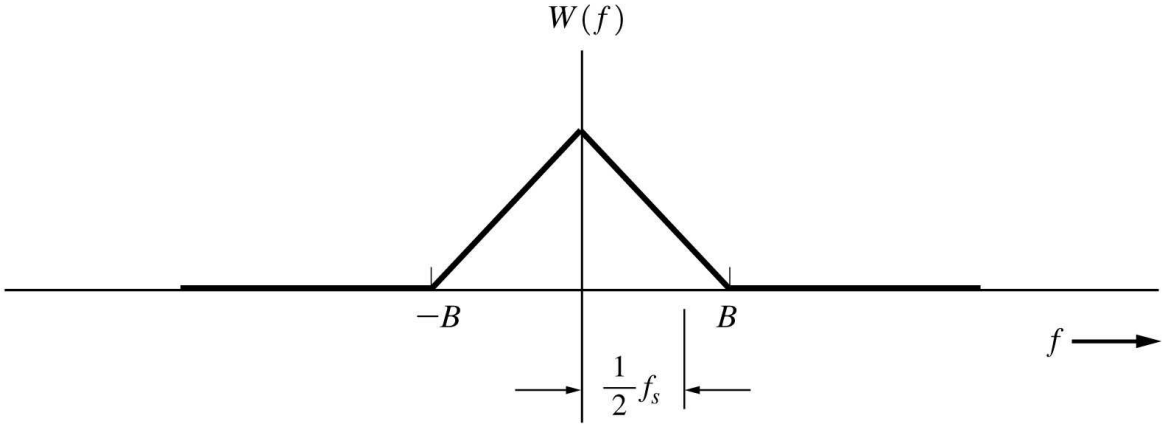
(a) Waveform and Its Spectrum



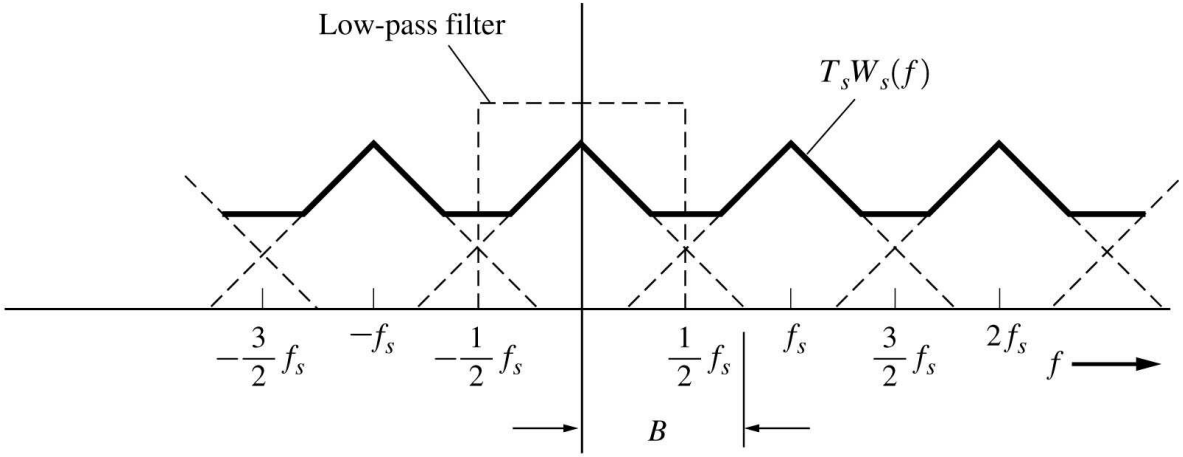
(b) Impulse Samples Waveform and Its Spectrum ($f_s > 2B$)

Impulse Sampling (5)

- If $f_s < 2B$, the signal is undersampled and the replicas of $W(f)$ will overlap resulting in aliasing.
- In this case, sampling introduces distortions to the signal.

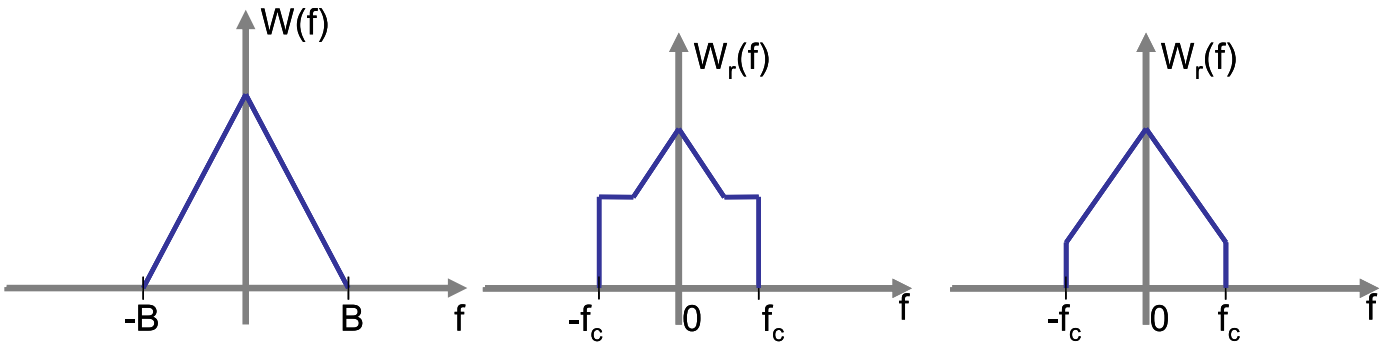


(a) Spectrum of Unsampled Waveform



Impulse Sampling (6)

- If the signal must be sampled at a frequency f_s, distortions can be reduced by prefiltering the signal before sampling.
  - The cutoff frequency of the lowpass prefilter must verify $f_c = f_s/2$.
- The prefilter still introduces distortions on the recovered signal since it chops off the spectrum of the original signal $w(t)$ above $f_s/2$.
- However, from the next figure we can observe that the distortions introduced by prefiltering are one-half the distortions caused by aliasing.



Spectrum of the signal before sampling

Reconstructed spectrum when aliasing occurs

Reconstructed spectrum with prefiltering

Bandwidth of Signals (1)

In engineering definitions, the bandwidth is taken to be the width of a positive frequency band.

- Note that we are dealing with real signals and the magnitude spectra of these signals are even functions.

The bandwidth would be:

$$B = f_2 - f_1$$

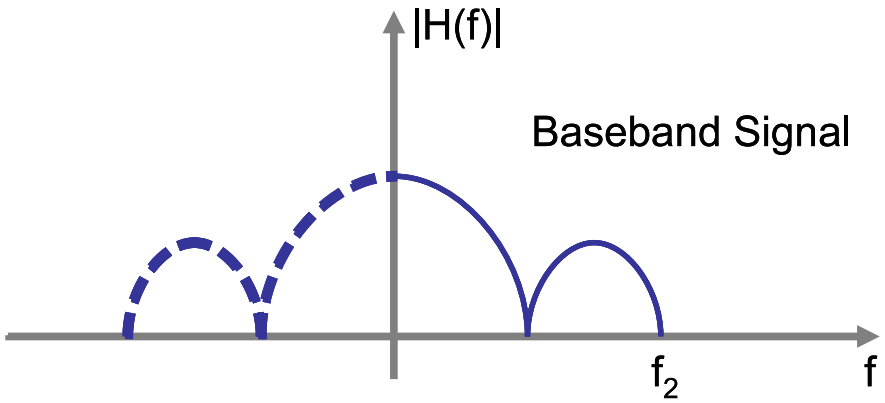
where $f_2 > f_1 \geq 0$ and f_1 and f_2 are determined by the particular definition that is used.

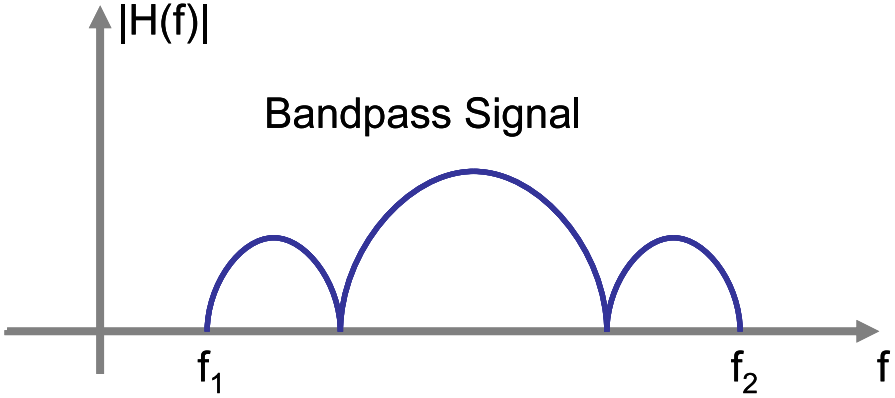
- For baseband signals: $f_1 = 0$ since the spectrum extends down to the DC ($f = 0$).
- For bandpass signals: $f_1 > 0$ and the band $f_1 \leq f \leq f_2$ encompasses the carrier frequency f_c.

Bandwidth of Signals (2)

Absolute Bandwidth: is $f_2 - f_1$ where the spectrum is zero outside the interval $[f_1 \quad f_2]$ (along the positive frequency axis).

- For the baseband signal sketched below: $B = f_2$.
- For the passband signal sketched below: $B = f_2 - f_1$.

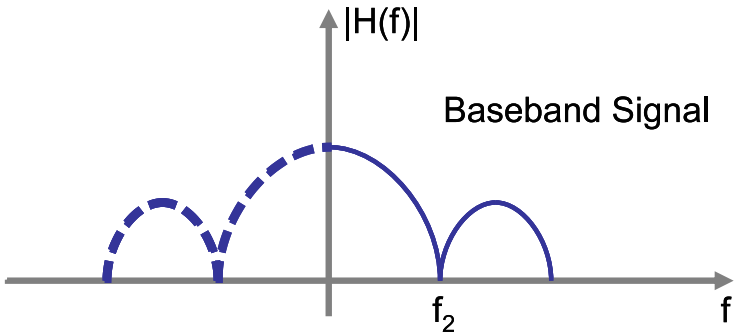


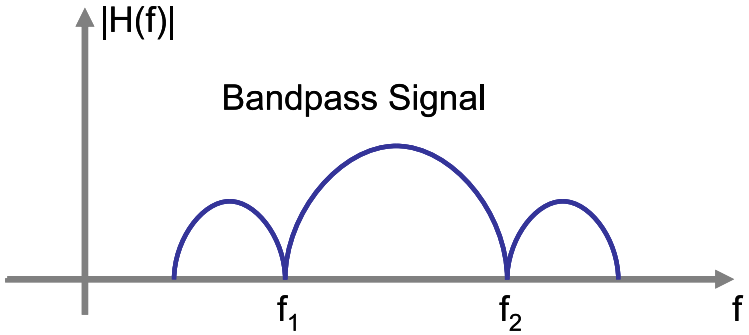


Bandwidth of Signals (3)

Null-to-null Bandwidth: Designate by f_0 the frequency where the magnitude spectrum is maximum:

- For baseband signals: $B = f_2$ where f_2 is the first null in the magnitude spectrum above f_0 (along positive frequencies).
- For passband signals: $B = f_2 - f_1$ where f_2 (resp. f_1) is the first null in the magnitude spectrum above (resp. below) f_0 (along positive frequencies).





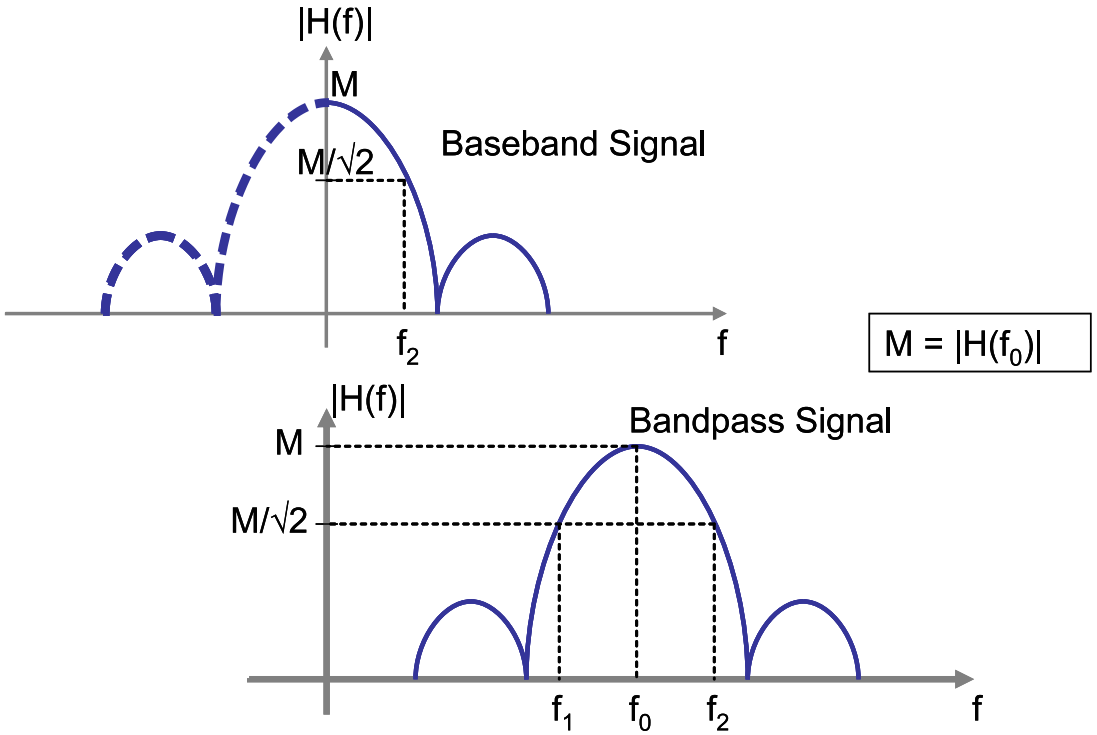
Bandwidth of Signals (4)

3-dB Bandwidth (or half-power bandwidth): Designate by f_0 the frequency where the magnitude spectrum is maximum:

- For passband signals: $B = f_2 - f_1$ where $f_1 \leq f_0 \leq f_2$ and:

$$|H(f)| \geq \frac{1}{\sqrt{2}}|H(f_0)| \quad ; f \in [f_1 \quad f_2]$$

- For baseband signals: the same definition holds with $f_1 = 0$.



Bandwidth of Signals (5)

- Note that at f_1 (and f_2):

$$|H(f_1)| = \frac{1}{\sqrt{2}}|H(f_0)|$$
$$\Rightarrow \frac{|H(f_1)|^2}{|H(f_0)|^2} = \frac{1}{2}$$

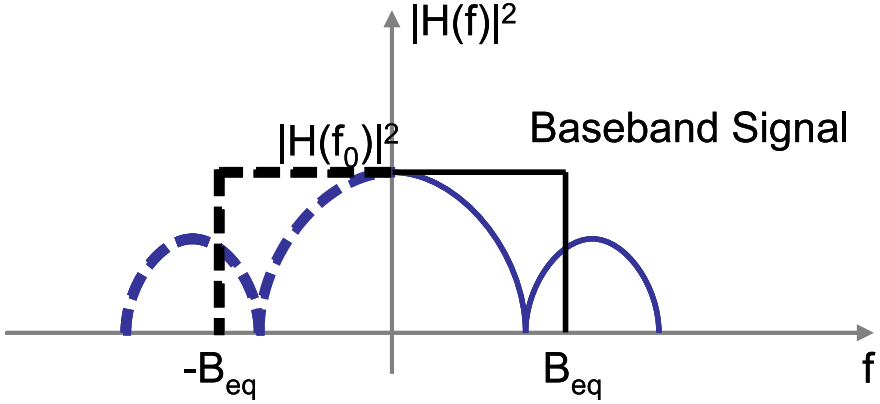
- In decibels:

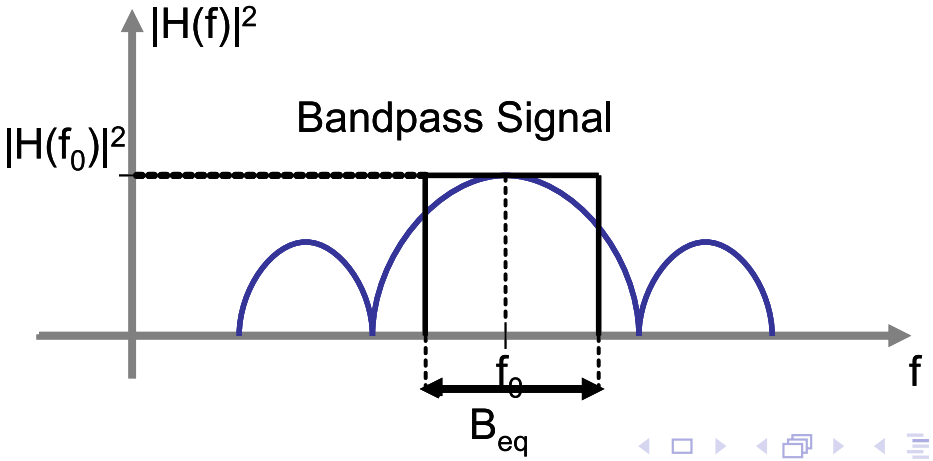
$$10\log_{10}\frac{|H(f_1)|^2}{|H(f_0)|^2} = -3 \text{ dB}$$

- In other words, $|H(f)|^2$ drops to half its maximum value (3 dB) at f_1 and f_2.

Bandwidth of Signals (6)

Equivalent noise Bandwidth: Designate by f_0 the frequency where the magnitude spectrum is maximum. The equivalent noise bandwidth is the width of a fictitious rectangular spectrum such that the power in that rectangular band is equal to the power associated with the actual spectrum (at positive frequencies).





Bandwidth of Signals (7)

- The power in the equivalent rectangular band is proportional to (at positive frequencies):

$$\text{Equivalent Power} = B_{eq}\,|H(f_0)|^2$$

- The actual power at positive frequencies is proportional to:

$$\text{Actual Power} = \int_0^{+\infty} |H(f)|^2\,\mathrm{d}f$$

- From the above equations, the equivalent noise bandwidth is:

$$B_{eq} = \frac{1}{|H(f_0)|^2}\int_0^{+\infty} |H(f)|^2\,\mathrm{d}f$$